

Amoeba: A More Flexible RLWE-based KEM

Contents

1	Description of Amoeba algorithm	2
1.1	Overview of Amoeba	2
1.2	Notations	2
1.3	IND-CPA Secure Amoeba.PKE	2
1.4	IND-CCA2 Amoeba.KEM	7
1.5	Parameter Sets	8
2	Design Rationale	8
2.1	RLWE Problem and Error Distributions	8
2.2	Ring Multiplication	10
2.2.1	Nussbaumer’s Trick	10
2.2.2	The NTT Trick	11
2.3	Analysis of Decoding Failure Rate	13
2.3.1	Noise Estimation	13
2.3.2	Selection of Error-correcting Codes	15
2.4	Criteria for Parameter Selection	16
2.5	Differences from Existing Schemes	16
3	Security Statement and Analysis	17
3.1	Theoretical Security Proof	17
3.1.1	Security Proof of Amoeba.PKE	18
3.1.2	Security Proof of Amoeba.KEM	20
3.2	Practical Security	21
3.2.1	Primal Attack	22
3.2.2	Dual Attack	22
3.2.3	Quantum Primal Attack	23
3.2.4	Quantum Dual Attack	24
3.2.5	Decryption Failure Attack	25
3.2.6	Practical Security of Amoeba	25
4	Performance Evaluation	27
4.1	Implementation details	27
4.2	Evaluation of our implementation	28
5	Feature Statements	29
	References	29

1 Description of Amoeba algorithm

1.1 Overview of Amoeba

The learning-with-errors (LWE) problem has become one of the most important parts in the design of Lattice-based post-quantum cryptography (PQC). RLWE-based schemes have attracted significant attention due to their relatively compact public key sizes and fast arithmetic operations [1, 2]. Most existing RLWE-based key encapsulation mechanisms (KEMs) are restricted to power-of-two cyclotomic rings to be compatible with the number theoretic transform (NTT), which causes a jump in the security strength.

Our scheme follows the standard RLWE-based KEM design paradigm, and adopts a more general cyclotomic ring of the form $\mathbb{Z}_q[X]/(X^{kn} - X^{kn/2} + 1)$ in order to eliminate the security level jump problem. At the same time, it maintains the performance advantages brought by the ring structure. We first propose an IND-CPA secure public-key encryption (PKE) scheme, and provide five sets of parameters which achieve security levels of 128, 192, 256, 384, and 512-bit against classical lattice attacks, and 80, 128 and 256-bit against quantum attacks respectively. By means of the well-known Fujisaki-Okamoto (FO) transform, an IND-CCA2 secure KEM scheme with corresponding security strengths can be obtained. The core technical feature of this scheme is to incorporate Nussbaumer's trick to realize a reusable NTT module over a common sub-ring $\mathbb{Z}_q[Y]/(Y^n - Y^{n/2} + 1)$ among different recommended parameter sets. This is also the origin of the scheme's name **Amoeba**. In addition, we use high-rate Hamming codes to strengthen the error-correcting code inherent in LWE-based PKE, reducing the decryption failure rate (DFR) to a negligible level.

1.2 Notations

Table 1 provides the basic mathematical notations used in this document and their explanations. Meanwhile, we list some commonly used abbreviations in Table 2 for readers' reference.

1.3 IND-CPA Secure Amoeba.PKE

In this section, we design a novel RLWE-based public-key encryption scheme named Amoeba.PKE, and the design details are as follows.

Cyclotomic Ring. Choose an integer $m = 2^s 3^t$, $s, t \geq 1$ and an appropriate prime q , satisfying $m \mid (q - 1)$. Let $n = \phi(m) = m/3$ and $k \in \mathbb{Z}^+$ be an integer related to the security level. Define $\mathcal{R}_q = \mathbb{Z}_q[X]/(\Phi_{km}(X))$. According to Section 2.2.1, there exists the ring isomorphism $\mathcal{R}_q \cong \mathbb{A}_q[X]/(X^k - Y)$, where $\mathbb{A}_q = \mathbb{Z}_q[Y]/(Y^n - Y^{n/2} + 1)$ and $Y = X^k \in \mathbb{A}_q$. This isomorphism and its inverse are defined as $\text{Nuss}(\cdot)$ and $\text{InvNuss}(\cdot)$, respectively.

Table 1: Explanation of mathematical notations

Notation	Description
Bold lower-case letters	Represent vectors. Unless otherwise specified, all vectors are assumed to be row ones.
Bold upper-case letters	Represent matrices.
$\mathbb{Z}$	The ring of integers.
$\mathbb{Q}$	The field of rational numbers.
$\mathbb{R}$	The field of real numbers.
$\mathbb{C}$	The field of complex numbers.
$\mathbb{Z}_q := \mathbb{Z}/q\mathbb{Z}$	The residue system modulo $q \in \mathbb{Z}^+$, where the coset representatives range from $[-q/2, q/2) \cap \mathbb{Z}$.
$[n]$	The set $\{0, 1, \dots, n-1\}$, where $n \in \mathbb{Z}^+$ is a positive integer.
$\mathcal{U}(S)$	The uniform distribution on a finite set S .
$s \leftarrow \mathcal{D}(S)$	Sample from the set S according to the distribution $\mathcal{D}$. For the case of uniform distribution, it is abbreviated as $s \leftarrow S$.
$\mathbf{v} \leftarrow \mathcal{D}^n(S)$	Component-wise independent sampling.
$\phi(m)$	Euler's totient function.
$\Phi_m(X)$	The m -th cyclotomic polynomial of degree $\phi(m)$.
$\zeta_m \in \mathbb{F}$	The m -th primitive root of unity in a certain field $\mathbb{F}$.
$\mathcal{R} := \mathbb{Z}[\zeta_m]$ $\cong \mathbb{Z}[X]/(\Phi_m(X))$	The m -th cyclotomic ring.
$\mathcal{K} := \mathbb{Q}(\zeta_m)$ $\cong \mathbb{Q}[X]/(\Phi_m(X))$	The m -th cyclotomic field.
$\lambda \in \mathbb{R}$	The security-quantifying parameter.

Polynomial Multiplication. Since k is a small integer, the multiplication operation in $\mathbb{A}_q[X]/(X^k - Y)$ can be rapidly computed using the schoolbook algorithm, where the basic operations are addition and multiplication in the ring $\mathbb{A}_q$. Due to $m \mid (q-1)$, there exists an m -th primitive root of unity ζ_m in $\mathbb{Z}_q$. Based on the results from Section 2.2.2, we have the following isomorphisms:

$$\mathbb{A}_q \cong \mathbb{Z}_q[X]/(Y^{n/2} - \zeta_6) \times \mathbb{Z}_q[Y]/(Y^{n/2} - \zeta_6^5), \quad (1)$$

$$\mathbb{Z}_q[Y]/(Y^{n/2} - \zeta_6) \cong \prod_{i=0}^{n/2-1} \mathbb{Z}_q[Y]/(Y - \zeta_m^{1+6i}) \cong \mathbb{Z}_q^{n/2}, \quad (2)$$

$$\mathbb{Z}_q[Y]/(Y^{n/2} - \zeta_6^5) \cong \prod_{i=0}^{n/2-1} \mathbb{Z}_q[Y]/(Y - \zeta_m^{5+6i}) \cong \mathbb{Z}_q^{n/2}. \quad (3)$$

Using the techniques from [8], the cost of each butterfly operation for the ring isomorphism in Equation 1 is one multiplication, addition, and subtraction in $\mathbb{Z}_q$, which is exactly the same as the cost of a radix-2 butterfly. The two smaller polynomial rings

Table 2: List of abbreviations and their full expressions

Abbreviation	Full expression
PKE	Public-key Encryption
KEM	Key Encapsulation Mechanism
LWE	Learning with Errors
RLWE	Ring-LWE
MLWE	Module-LWE
CVP	Closest Vector Problem
SVP	Shortest Vector Problem
IND-CPA	Indistinguishability under Chosen-plaintext Attack
IND-CCA	Indistinguishability under Chosen-ciphertext Attack
ROM	Random Oracle Model
QROM	Quantum ROM
PRG	Pseudo-random Generator
PRF	Pseudo-random Function
XOF	Extendable Output Function
KDF	Key-derivation Function
FFT	Fast Fourier Transform
NTT	Number Theoretic Transform
DFR	Decryption Failure Rate
ECC	Error-correcting Code

sequentially perform $t - 1$ layers of radix-3 butterflies and $s - 1$ layers of radix-2 butterflies. The above isomorphism process are denoted as $\text{NTT}(\cdot)$, and its inverse is denoted as $\text{InvNTT}(\cdot)$. Given a ring element $a \in \mathcal{R}_q$, we denote its NTT form as $\hat{a}$. Algorithm 1 and Algorithm 2 describe this conversion process and its inverse, respectively.

There are two organizational forms for $\hat{a}$. The first is $\hat{a} \in (\mathbb{Z}_q)^n[X]/(X^k - Y)$, which represents viewing $a \in \mathcal{R}_q$ as a single polynomial $a' \in \mathbb{A}_q[X]/(X^k - Y)$, with each coefficient $a'_i \in \mathbb{A}_q$ stored in NTT form. The second is $\hat{a} \in (\mathbb{Z}_q[X]/(X^k - Y))^n$, which signifies a vector of n polynomials in the ring $\mathbb{Z}_q[X]/(X^k - Y)$, denoted as $\hat{a} = (\hat{a}^{(0)}, \dots, \hat{a}^{(n-1)})$. Based on the isomorphism induced by Nussbaumer's trick and NTT, the product of two polynomials $a, b \in \mathcal{R}_q$ is equivalent to the component-wise multiplication of the small polynomials that make up the vectors $\hat{a}$ and $\hat{b}$, in the sense of isomorphism. We denote this multiplication operation as $\hat{a} \circ \hat{b}$, described in Algorithm 3.

Sampling of Private Keys and Noise. Since km is even, according to the main reduction theorem in [9], we have $\Theta_{km}(X) = X^{km'} + 1$, where $m' = m/2$. For sampling efficiency, we use the central binomial distribution instead of the discrete Gaussian distribution. Let β_η denote the CBD with parameter η . First, we sample $e' \leftarrow \beta_\eta^{km'}$ over $\mathbb{Z}[X]/(\Theta_{km}(X))$, and then compute $e \leftarrow e' \pmod{\Phi_{km}(X)}$ as the output. Since $\Phi_{km}(X)$ is a sparse polynomial, e can be obtained efficiently from the coefficient representation

Algorithm 1 NussNTT

Input: $a \in \mathbb{Z}_q[X]/(X^{kn} - X^{kn/2} + 1)$ **Output:** $\hat{a} \in (\mathbb{Z}_q[X]/(X^k - Y))^n$

- 1: $a' := \text{Nuss}(a) \in \mathbb{A}_q[X]/(X^k - Y)$ $\triangleright a' = \sum_{i=0}^{k-1} a'_i X^i$
 - 2: **for** $i = 0, \dots, k-1$ **do**
 - 3: $\hat{a}_i := \text{NTT}(a'_i) \in \mathbb{Z}_q^n$
 - 4: **end for**
 - 5: **return** $\hat{a}$
-

Algorithm 2 InvNussNTT

Input: $\hat{a} \in (\mathbb{Z}_q[X]/(X^k - Y))^n$ **Output:** $a \in \mathbb{Z}_q[X]/(X^{kn} - X^{kn/2} + 1)$

- 1: **for** $i = 0, \dots, k-1$ **do**
 - 2: $a'_i := \text{InvNTT}(\hat{a}_i) \in \mathbb{A}_q$ $\triangleright a' = \sum_{i=0}^{k-1} a'_i X^i$
 - 3: **end for**
 - 4: $a := \text{InvNuss}(a')$
 - 5: **return** a
-

Algorithm 3 NussMul

Input: $\hat{a}, \hat{b} \in (\mathbb{Z}_q[X]/(X^k - Y))^n$ $\triangleright \hat{a} = (\hat{a}^{(0)}, \dots, \hat{a}^{(n-1)}), \hat{b} = (\hat{b}^{(0)}, \dots, \hat{b}^{(n-1)})$ **Output:** $\hat{c} = \text{NussNTT}(a \cdot b) \in (\mathbb{Z}_q[X]/(X^k - Y))^n$

- 1: **for** $j = 0, \dots, n-1$ **do**
 - 2: $\hat{c}^{(j)} := \hat{a}^{(j)} \cdot \hat{b}^{(j)} \bmod (q, X^k - Y)$ $\triangleright$ Schoolbook multiplication
 - 3: **end for**
 - 4: **return** $\hat{c}$ $\triangleright \hat{c} = (\hat{c}^{(0)}, \dots, \hat{c}^{(n-1)})$
-

Algorithm 4 Sampler

Input: $\eta \in \mathbb{Z}$ **Output:** $e \in \mathcal{R}$

- 1: $e' \leftarrow \beta_\eta^{km'}$ $\triangleright e' = \sum_{i=0}^{km'-1} e'_i X^i \in \mathbb{Z}[X]/(\Theta_{km}(X))$
 - 2: $e := e' \pmod{\Phi_{km}(X)}$
 - 3: **return** e
-

of e' through a simple linear transformation. Algorithm 4 describes our sampling procedure, and we denote it as $e \leftarrow \psi_\eta(\rho)$, where $\rho \in \{0, 1\}^*$ is the random seed used for β_η . To meet the 512-bit security level, Amoeba samples 512-bit seeds ρ, σ, r in Algorithm 5 and Algorithm 6 from a true random number generator.

Compression of Public Keys and Ciphertexts. To reduce the size of public keys, we utilize an XOF along with a random seed ρ to reconstruct the polynomial $\hat{a} := \text{Parse}(\text{XOF}(\rho))$, keeping only the seed as part of the public key. We set parameters $d_0, d_1, d_2 \in \mathbb{Z}^+$, which denote the coefficients scaling down from $\mathbb{Z}_q$ to $\mathbb{Z}_{2^{d_i}}$ for the second

Algorithm 5 PKE.KeyGen

Input: 1^κ **Output:** sk, pk

- 1: $\rho, \sigma \leftarrow \{0, 1\}^{512}$
 - 2: $\hat{a} := \text{Parse}(\text{XOF}(\rho))$
 - 3: $s \leftarrow \psi_{\eta_1}(\text{PRF}(\sigma, 0x00))$,
 - 4: $e \leftarrow \psi_{\eta_2}(\text{PRF}(\sigma, 0x01))$
 - 5: $\hat{s} := \text{NussNTT}(s)$
 - 6: $b := \text{InvNussNTT}(\hat{s} \circ \hat{a}) + e$
 - 7: $t := \text{Pack}(\text{ModDown}(b, d_0))$
 - 8: **return** $sk := \hat{s}, pk := (\rho, t)$
-

Algorithm 6 PKE.Enc

Input: $pk = (\rho, t)$, $m \in \{0, 1\}^{512}$, $r \in \{0, 1\}^{512}$ **Output:** ct

- 1: $\hat{a} := \text{Parse}(\text{XOF}(\rho))$
 - 2: $b' := \text{ModUp}(\text{Unpack}(t), d_0)$
 - 3: $s' \leftarrow \psi_{\eta_1}(\text{PRF}(r, 0x02))$
 - 4: $e_1 \leftarrow \psi_{\eta_2}(\text{PRF}(r, 0x04))$
 - 5: $e_2 \leftarrow \psi_{\eta_2}(\text{PRF}(r, 0x08))$
 - 6: $\hat{s}' := \text{NussNTT}(s')$
 - 7: $\hat{b}' := \text{NussNTT}(b')$
 - 8: $u := \text{InvNussNTT}(\hat{a} \circ \hat{s}') + e_1$
 - 9: $v := \text{InvNussNTT}(\hat{b}' \circ \hat{s}') + e_2 + \text{Encode}(m)$
 - 10: $c_1 := \text{Pack}(\text{ModDown}(u, d_1))$
 - 11: $c_2 := \text{Pack}(\text{ModDown}(\text{Trunc}(v), d_2))$
 - 12: **return** $ct := (c_1, c_2)$
-

component b of the public key, the first component c_1 and the second component c_2 of the ciphertext, respectively. This kind of technique is known as modulus switching in homomorphic encryption. The compression and decompression procedures are denoted as $\text{ModDown} : a \in \mathbb{Z}_q \mapsto \lfloor 2^d a / q \rfloor \in [2^d]$ and $\text{ModUp} : a' \in [2^d] \mapsto \lfloor qa' / 2^d \rfloor \in \mathbb{Z}_q$, with parameter $d \in \mathbb{Z}^+$.

Encoding of Messages. Our ultimate design goal is an IND-CCA KEM that attains security levels of 128, 256, and 512 bits respectively. Therefore, we fix the message length at 512 bits across all security levels to achieve a unified interface. The length of the second component of the ciphertext for Amoeba is greater than 512, enabling the use of error-correcting codes for 512-bit messages. We defer the determination of the adopted error-correcting codes to Section 2.3.2, after the noise analysis. The encoding process, denoted as $\text{Encode} : \{0, 1\}^{512} \rightarrow \mathbb{Z}_q^\ell$, encodes a message into a vector of length

Algorithm 7 PKE.Dec

Input: $sk = \hat{s}$, $ct = (c_1, c_2)$ **Output:** $m' \in \{0, 1\}^{512}$

- 1: $u' := \text{ModUp}(\text{Unpack}(c_1), d_1)$
 - 2: $v' := \text{ModUp}(\text{Unpack}(c_2), d_2)$
 - 3: $\hat{u}' := \text{NussNTT}(u')$
 - 4: $w := \text{InvNussNTT}(\hat{s} \circ \hat{u}')$
 - 5: $m' := \text{Decode}(v' - \text{Trunc}(w))$
 - 6: **return** m'
-

$\ell \leq kn$. The corresponding decoding process is denoted as $\text{Decode} : \mathbb{Z}_q^\ell \rightarrow \{0, 1\}^{512}$. If the codeword length is less than the dimension of $\mathcal{R}_q$, then the $kn - \ell$ coefficients of c_2 that do not store codeword information can be discarded. This process is denoted as $\text{Trunc} : \mathcal{R}_q \rightarrow \mathbb{Z}_q^\ell$.

1.4 IND-CCA2 Amoeba.KEM

We use the FO transformation [6, 7] to obtain an IND-CCA-secure Amoeba.KEM under quantum random oracle model (QROM). The FO transformation is the main technology for enhancing PKEs from IND-CPA security to IND-CCA security. We employ $\text{FO}_{\text{ID}(pk), m}^\perp$ proposed by [10], which offers a tight reduction in the multi-user setting.

Let $\text{ID} : \text{PK} \rightarrow \{0, 1\}^{512}$ be a function that maps public keys to fixed-length random strings with sufficient min-entropy. The encapsulation process is as follows. First, a random number $m \in \{0, 1\}^{512}$ is sampled, and a hash function $G : \{0, 1\}^* \rightarrow \{0, 1\}^{1024}$ is used to generate $(\bar{K}, r) := G(\text{ID}(pk), m)$. The pre-key $\bar{K}$ is used for the derivation of the shared key, and the random tape r is used to provide randomness for the encryption of PKE. Subsequently, the encryption of PKE is called to generate the ciphertext of $ct := \text{Enc}(pk, m; r)$, and a key-derivation function $\text{KDF} : \{0, 1\}^* \rightarrow \{0, 1\}^*$ is used to generate the shared key $K := \text{KDF}(\bar{K}, ct)$. The decapsulation is straightforward. When the verification fails in the re-encryption process, it returns a fake key $K' := \text{KDF}(z, ct)$, where z is a fixed secret string.

The public key and ciphertext of Amoeba.KEM are the same as those of Amoeba.PKE, and their sizes are as follows:

$$|\text{KEM.PK}| = |\rho| + \left\lceil \frac{d_0 \cdot k \cdot n}{8} \right\rceil, \quad (4)$$

$$|\text{KEM.CT}| = \left\lceil \frac{d_1 \cdot k \cdot n}{8} \right\rceil + \left\lceil \frac{d_2 \cdot \ell}{8} \right\rceil. \quad (5)$$

Since the private key of Amoeba.KEM is only stored locally, apart from the private key of Amoeba.PKE and the secret z used to generate the fake key, incorporating the complete public key of Amoeba.PKE into its composition can reduce the computational

overhead of decapsulation.

$$|\text{KEM.SK}| = \left\lceil \frac{[\log q] \cdot k \cdot n}{8} \right\rceil + |z| + |\text{KEM.PK}|. \quad (6)$$

We can even further expand the seed ρ into ring elements and then store them in the private key of Amoeba.KEM, so as to optimize the re-encryption in decapsulation.

According to the solicitation requirements, we use SM3 to instantiate Hash, XOF and PRF. These symmetric primitives are plugins and will need to be replaced with algorithms that meet at least a 512-bit classical security strength and a 256-bit quantum security strength in the future. To resist collision attacks, the output length of the Hash should be at least 1024 bits, which exactly meets the instantiation requirements of function G.

In our PKE scheme, PRF is used to generate s, e, s', e_1 , and e_2 . We employ oracle-cloning techniques [11, 12], using prefix-free labels 0x00, 0x01, 0x02, 0x04, and 0x08 to differentiate them, ensuring these five random oracles are independent of each other. The input lengths of XOF and KDF are different, allowing them to be treated as independent random oracles.

1.5 Parameter Sets

We present five recommended sets of parameters in Table 3, denoted as Amoeba-X. The suffix X is the ring dimension of the RLWE problem. In addition to the parameter sets with security strengths of 128, 256, and 512 bits, we also provide parameter sets achieving security strengths of 192 and 384 bits, which reflects the flexibility in the selection of parameter sets in this scheme.

Table 4 shows the sizes of the payloads and decryption failure rate for these parameter sets. The private key can be generated online through a seed, so that it can be compressed to several tens of bytes. Amoeba-1728 and Amoeba-2304 use the same noise parameters as Amoeba-1152. As the ring dimension increases, more cumulative noise leads to an increased error rate on each coefficient. If a lower DFR is desired, the Hamming code should be replaced with a more capable error-correcting code, at the cost of an increase in the size of the second component of the ciphertext. Alternatively, when ensuring that the security strength attains the required standard, one may opt to employ smaller non-integer parameters of the central binomial distribution. Subsequent practical security evaluations indicate that the security strength of these two parameter sets is several tens of bits higher than required.

2 Design Rationale

2.1 RLWE Problem and Error Distributions

The formal definition of the RLWE problem is provided by [1]. For the tightness of the reduction and tolerance to errors, they used complex algebraic structures such as

Table 3: Recommended parameter sets of Amoeba

Parameters	q	n	k	ℓ	η_1	η_2	(d_0, d_1, d_2)
Amoeba-576	3457	288	2	523	2	2	(10,10,5)
Amoeba-864	3457	288	3	523	1	2	(10,11,6)
Amoeba-1152	3457	288	4	523	1	1	(11,10,6)
Amoeba-1728	3457	288	6	523	1	1	(11,11,6)
Amoeba-2304	3457	288	8	523	1	1	(12,12,7)

Table 4: Payload size of Amoeba.KEM in bytes and decryption failure rate

Parameters	Private Key	Public Key	Ciphertext	DFR
Amoeba-576	1712	784	1047	$2^{-133.3}$
Amoeba-864	2504	1144	1581	$2^{-208.6}$
Amoeba-1152	3440	1648	1833	$2^{-267.9}$
Amoeba-1728	5096	2440	2769	$2^{-229.3}$
Amoeba-2304	7040	3520	3914	$2^{-205.3}$

fractional ideals and dual ideals. In this work, we use the non-dual version of the RLWE problem, which is equivalent to the dual form RLWE through a twist deformation [9].

Definition 2.1 (Non-dual form RLWE). *Let $s \in \mathcal{R}_q$ be a secret element, and let ψ be an error distribution over the cyclotomic field $\mathcal{K}$. Uniformly sample $a \leftarrow \mathcal{R}_q$, sample $e \leftarrow \psi$, and output $(a, b = a \cdot s + e \pmod{q\mathcal{R}}) \in \mathcal{R}_q \times \mathcal{K}/q\mathcal{R}$. The output distribution is called the RLWE distribution and is denoted by $\mathcal{A}_{s,\psi}$.*

- **Search-RLWE problem.** *Let Ψ be a family of error distributions over $\mathcal{K}$. For any $s \in \mathcal{R}_q$ and $\psi \in \Psi$, given an arbitrary number of independent and identically distributed RLWE samples $(a_i, b_i) \leftarrow \mathcal{A}_{s,\psi}$, find s with non-negligible advantage.*
- **Decision-RLWE problem.** *Let $\mathcal{P}$ be a distribution over a family of error distributions over $\mathcal{K}$. For uniformly chosen $(s, \psi) \leftarrow \mathcal{U}(\mathcal{R}_q) \times \mathcal{P}$, given an arbitrary number of independent and identically distributed samples (a_i, b_i) , distinguish with non-negligible advantage whether they are from the RLWE distribution $\mathcal{A}_{s,\psi}$ or the uniform distribution $\mathcal{U}(\mathcal{R}_q \times \mathcal{K}/q\mathcal{R})$.*

The error distribution ψ should be an elliptic Gaussian distribution over the m -th cyclotomic field $\mathcal{K}$, according to the main reduction theorem from [1]. For the special case where $m = 2^s, s \geq 1$, the error distribution ψ is precisely a spherical Gaussian distribution, allowing us to independently sample each coefficient of the error e from a one-dimensional Gaussian distribution. However, in this work, a more general cyclotomic ring is used. If we still sample each coefficient independently, the geometric shape of the noise might leak private key information.

By twisting the decoding basis of $\mathcal{R}^\vee$ [17], the correct Gaussian distribution can be sampled. A simpler conversion technique proposed by [9] does not require complex algebraic structures such as fractional ideals and their duals. Define the polynomial

$\Theta_m(X)$ as

$$\Theta_m(X) = \begin{cases} X^{m/2} + 1, & m \text{ is even,} \\ X^m - 1, & m \text{ is odd.} \end{cases} \quad (7)$$

Let $m' = \deg \Theta_m(X)$. First, sample the error from a spherical Gaussian distribution $\mathcal{D}_w^{m'}$ over the extension ring $\mathbb{Q}[X]/(\Theta_m(X))$ of $\mathcal{K}$, where $w \in \mathbb{R}$ is the Gaussian parameter. Then, simply modulo the polynomial $\Phi_m(X)$ to move it back to $\mathcal{K}$. This conversion only has a small loss in the tightness of reduction.

Theorem 2.1 ([9]). *Let $\alpha \in (0, 1)$ be a real number, and $q = q(n)$ be a polynomial-size prime number such that $\alpha q > \omega(\sqrt{\log n})$ and $q \equiv 1 \pmod{m}$. For any integer $k \geq 1$, suppose there exists a probabilistic polynomial-time algorithm that, for s uniformly chosen over $\mathcal{R}_q$, given at most k independent samples, can distinguish with non-negligible advantage whether the samples come from $\mathcal{A}_{s,\psi}$ or from $\mathcal{U}(\mathcal{R}_q \times \mathcal{R}_q)$, where the noise distribution ψ is sampled as $e = \lfloor e' \bmod \Phi_m(X) \rfloor \in \mathcal{R}$, with $e' \in \mathbb{Q}[X]/(\Theta_m(X))$ following the spherical Gaussian distribution $\mathcal{D}_{\sqrt{m'} \cdot \alpha q \cdot (nk/\log(nk))^{1/4}}^{m'}$. Then there exists a polynomial-time quantum algorithm that can solve the approximate SVP problem on ideal lattices in $\mathcal{K}$, within an approximation factor $\tilde{O}(\sqrt{m}/\alpha)$.*

Although this theorem requires $m \mid (q-1)$, it can be proven that the pseudorandomness of RLWE holds for any modulus [18], using modulus-switching technique. For the spherical Gaussian sampler on the extension ring $\mathbb{Q}[X]/(\Theta_m(X))$, a widely used strategy for sampling efficiency is to replace the Gaussian distribution with a central binomial distribution (CBD), which does not lead to significant security loss.

2.2 Ring Multiplication

2.2.1 Nussbaumer's Trick

Nussbaumer's trick was originally used to construct some suitable roots of unity for those convolution rings which do not naturally support the NTT algorithm, but in this work it is used for the purpose of code reuse for multiplication in cyclotomic rings, and we make a small modification.

The literature [19] described Nussbaumer's trick as a ring isomorphism. Given the polynomial ring $\mathbb{Z}_q[X]/(X^{kn} + 1)$, where n is a power of two and q, k are integers. Let $Y = X^k$. There exists a ring isomorphism:

$$\mathbb{Z}_q[X]/(X^{kn} + 1) \cong (\mathbb{Z}_q[Y]/(Y^n + 1)) [X]/(X^k - Y). \quad (8)$$

Precisely, it is a simple permutation of the polynomial coefficients, that is, $f(X) = \sum_{j=0}^{k-1} \left(\sum_{i=0}^{n-1} f_{ik+j} Y^i \right) X^j$. The range of the coefficients on the right side of Equation 8 is $\mathbb{A}_q = \mathbb{Z}_q[Y]/(Y^n + 1)$. Note that $Y \in \mathbb{A}_q$ is a $2n$ -th primitive root of unity. The idea behind Nussbaumer's trick is to lift the ring $\mathbb{A}_q[X]/(X^k - Y)$ to a larger ring $\mathbb{A}_q[X]/(X^{2n} - 1)$, employing Y as the primitive root to execute the NTT algorithm.

We adopt the same ring isomorphism over non-power-of-two cyclotomic rings. But instead of performing NTT on $\mathbb{A}_q[X]/(X^k - Y)$, we perform NTT on $\mathbb{A}_q$ itself. We focus on the multiplication in the ring $\mathcal{R}_q = \mathbb{Z}_q[X]/(\Phi_{km}(X))$, where $m = 2^s 3^t$, $s, t \geq 1$ being a fixed integer. By setting $Y = X^k$, there exists a ring isomorphism:

$$\mathcal{R}_q \cong \left(\mathbb{Z}_q[Y]/(Y^n - Y^{n/2} + 1) \right) [X]/(X^k - Y), \quad (9)$$

where $n = m/3$ and $Y^n - Y^{n/2} + 1$ is a fixed cyclotomic polynomial. Define $\mathbb{A}_q = \mathbb{Z}_q[Y]/(Y^n - Y^{n/2} + 1)$. As described in the following subsection, if $m \mid (q-1)$, we can perform the NTT algorithm on $\mathbb{A}_q$, which is fully reused for different values of k . Eventually, we achieve an isomorphism $\mathcal{R}_q \cong (\mathbb{Z}_q[X]/(X^k - Y))^n$, which is employed in the multiplication of polynomials within $\mathcal{R}_q$.

2.2.2 The NTT Trick

The general term for polynomial rings of the form $\mathbb{F}[X]/(X^N \pm 1)$ is (negacyclic) convolution rings, where $\mathbb{F}$ is a field. We also refer to polynomial rings of the form $\mathbb{F}[X]/(X^N - \zeta)$ as convolution rings, where $\zeta \in \mathbb{F}$ is any element.

The fast Fourier transform (FFT) algorithm can compute the isomorphism of convolution ring $\mathbb{C}[X]/(X^N - 1) \cong \prod_{j=0}^{N-1} \mathbb{C}[X]/(X - \exp(j \cdot 2\pi i/N))$ with a complexity of $O(N \log N)$. The number theoretic transform (NTT) is a simple analogue of the FFT in finite fields. The most common one is the radix-2 butterfly algorithm, which can be used for convolution rings of dimensions that are powers of two over finite fields whose characteristic is not 2. The authors of [2] introduced the sparse decomposition of arbitrary cyclotomic rings from the perspective of tensor decomposition of matrices. From the perspective of CRT, we extend the mixed-radix butterfly algorithm [20] from convolution rings to arbitrary cyclotomic rings.

Theorem 2.2. *Given an integer $m = \prod_{i=1}^t p_i^{e_i}$, where p_i are distinct primes and $e_i \geq 1$. Let q be a prime such that $m \mid (q-1)$. There is an algorithm that implements the forward and inverse transformations within the ring isomorphism $\mathbb{Z}_q[X]/(\Phi_m(X)) \cong \mathbb{Z}_q^{\phi(m)}$, with a computational complexity of*

$$O \left(\phi(m) \cdot \left(\phi \left(\prod_{i=1}^t p_i \right) + \sum_{i=1}^t (p_i - 1)(e_i - 1) \right) \right). \quad (10)$$

Decomposition of Cyclotomic Rings. For an integer $m = \prod_{i=1}^t p_i^{e_i}$, $e_i \geq 1$, the radical of m is defined as $\text{Rad}(m) = \prod_{i=1}^t p_i$. The m -th cyclotomic polynomial can be expressed as $\Phi_m(X) = \Phi_{\text{Rad}(m)}(X^{m/\text{Rad}(m)})$. With shorthand notations $m' = \text{Rad}(m)$ and $N = m/m'$, $\Phi_m(X)$ can be decomposed into a product of several coprime linear polynomials in the variable X^N ,

$$\Phi_m(X) = \Phi_{m'}(X^N) = \prod_{j \in \mathbb{Z}_{m'}^*} (X^N - w_{m'}^j), \quad (11)$$

where $w_{m'}$ is an abstract m' -th primitive root of unity in the algebraic closure. Note that $\phi(m') = n/N$, where $n = \phi(m)$. Let $\zeta_{m'}$ denote a primitive m' -th root of unity in the prime field $\mathbb{Z}_q$, where q is a prime satisfying $m' \mid (q-1)$. According to the CRT, there exists an isomorphism:

$$\mathbb{Z}_q[X]/(\Phi_m(X)) \cong \prod_{j \in \mathbb{Z}_{m'}^*} \mathbb{Z}_q[X]/(X^N - \zeta_{m'}^j). \quad (12)$$

Given any polynomial $g(X) = \sum_{i=0}^n g_i X^i \in \mathbb{Z}_q[X]/(\Phi_m(X))$, let $f_j = g \pmod{X^N - \zeta_{m'}^j}$. It is straightforward to compute

$$f_j(X) = \sum_{i=0}^{N-1} \left(g_i + \zeta_{m'}^j \cdot g_{i+N} + \cdots + \zeta_{m'}^{(\phi(m')-1)j} \cdot g_{i+(\phi(m')-1)N} \right) \cdot X^i. \quad (13)$$

One can notice that for any $i^* \in [N]$, the i^* -th coefficients of all the polynomials f_j are determined by $\phi(m')$ coefficients $g_{i^*+kN}, k \in [\phi(m')]$. This computational unit is formally very similar to the CT butterfly, and its computational cost is $\phi(m') \cdot (\phi(m') - 1)$ multiplications and $\phi(m') \cdot (\phi(m') - 1)$ additions over $\mathbb{Z}_q$. For this ring isomorphism, a total of N identical structural computing units are required, and the overall computational complexity is $N \cdot \phi(m') \cdot (\phi(m') - 1) = n \cdot (\phi(m') - 1)$. By adopting $m = 2^s 3^t, s, t \geq 1$, we have $m' = \text{Rad}(m) = 6$ and $\phi(m') = 2$, thus the overall computational complexity should be $O(n)$. An observation from [8] can be employed to enhance efficiency. Using the equation $\zeta_6^5 = 1 - \zeta_6$, one multiplication in the computational unit can be transformed into an addition. Then, the computational complexity becomes $n/2$ multiplications and $3n/2$ additions. In general, multiplication over $\mathbb{Z}_q$ is much slower than addition.

We use $f_j[i], j \in \mathbb{Z}_{m'}^*, i \in [N]$ to denote the coefficient of the X^i term in polynomial f_j . For the inverse transform, we first compute

$$\phi(m') \cdot g_{i+kN} = \sum_{j \in \mathbb{Z}_{m'}^*} \zeta_{m'}^{-jk} \cdot f_j[i], \forall i \in [N], \forall k \in [\phi(m')]. \quad (14)$$

Then, we multiply by a scale factor $\phi(m')^{-1} \pmod{q}$ to obtain the coefficients of the polynomial $g(x) \in \mathbb{Z}_q[X]/(\Phi_m(X))$. The complexity of the inverse transform is the same as that of the forward transform.

Butterfly Algorithm. Now the cyclotomic ring $\mathbb{Z}_q[X]/(\Phi_m(X))$ is decomposed into $\phi(m')$ convolution rings of the form $\mathbb{Z}_q[X]/(X^N - \zeta)$, where $\zeta = \zeta_{m'}^j$ is some m' -th primitive root of unity. For a prime $p \mid N$, assuming that there exists p -th roots of ζ in $\mathbb{Z}_q$, the polynomial $X^N - \zeta$ can be factored into p coprime polynomials. Specifically, $X^N - \zeta = \prod_{j=0}^{p-1} (X^{N/p} - \zeta_{m'p}^{j+km'})$, where $\{\zeta_{m'p}^{j+km'}\}_{k \in [p]}$ are all the p -th roots of ζ . Similar to Formula 12, the following isomorphism holds:

$$\mathbb{Z}_q[X]/(X^N - \zeta) \cong \prod_{k=0}^{p-1} \mathbb{Z}_q[X]/(X^{N/p} - \zeta_{m'p}^{j+km'}). \quad (15)$$

The corresponding mapping and its inverse are as follows:

$$\begin{aligned} h_k(X) &= \sum_{i=0}^{N/p-1} \left(\sum_{l=0}^{p-1} \zeta_{m'p}^{l(j+km')} \cdot f_{i+lN/p} \right) \cdot X^i, \\ p \cdot f_{i+lN/p} &= \sum_{k=0}^{p-1} \zeta_{m'p}^{-l(j+km')} \cdot h_k[i], \quad \forall i \in [N/p], \quad \forall l \in [p], \end{aligned} \quad (16)$$

where $f(X) = \sum_i f_i X^i \in \mathbb{Z}_q[X]/(X^N - \zeta)$ is an arbitrary polynomial, and $h_k = f \pmod{X^{N/p} - \zeta_{m'p}^{j+km'}}$.

Similar to the complexity analysis process in the previous section, the mapping described above uses the radix- p butterfly as its computational unit. The entire ring isomorphism requires N/p butterfly operations, with a total overhead of $N(p-1)$ multiplications and additions in $\mathbb{Z}_q$. For the special case of $p = 2$, the relation $\zeta_{2m'}^{j+m'} = -\zeta_{2m'}^j$ allows us to remove half of the multiplication operations, thus requiring only $N/2$ multiplications and N additions over $\mathbb{Z}_q$.

2.3 Analysis of Decoding Failure Rate

Based on the parameter selection strategy described above, we now provide a rigorous mathematical analysis of the decryption failure rate to ensure it is sufficiently low for the Fujisaki-Okamoto transform.

2.3.1 Noise Estimation

In our PKE construction, sources of noise include the secret $s, s' \leftarrow \psi_{\eta_1}$, the noise $e, e_1, e_2 \leftarrow \psi_{\eta_2}$, and the rounding error introduced by compression b, c_1, c_2 . Since we adopt a non-power-of-two cyclotomic ring, this leads to differences in noise distribution compared to the scenario of using power-of-two cyclotomic ring.

Lemma 2.1 ([13]). *The product of two variables in the cyclotomic ring satisfies the following properties.*

- Let $m = 2^s$ be an integer with $s \geq 1$. Given two random variables $a, b \in \mathbb{Z}[X]/(X^n + 1)$, where $n = m/2$, and their variances are σ_a^2 and σ_b^2 , respectively. The variance σ_c^2 of the product $c = ab$ satisfies $\sigma_c^2 \leq n\sigma_a^2\sigma_b^2$.
- Let $m = 2^s 3^t$ be an integer with $s, t \geq 1$. Given two random variables $a, b \in \mathbb{Z}[X]/(X^n - X^{n/2} + 1)$, where $n = m/3$, and their variances are σ_a^2 and σ_b^2 , respectively. The variance σ_c^2 of the product $c = ab$ satisfies $\sigma_c^2 \leq \frac{3n}{2}\sigma_a^2\sigma_b^2$.

Theorem 2.3. *Given a set of parameters for the PKE scheme described by algorithms 5, 6, and 7, namely, the cyclotomic index m , the modulus q , the central binomial distribution parameters η_1, η_2 , and the compression parameters d_0, d_1, d_2 . Under this set of parameters, the variance σ_E^2 of the coefficients of decryption noise satisfies:*

$$\sigma_E^2 \leq (m\eta_1 + 1)\eta_2 + \frac{m\eta_1 (\lceil q/2^{d_0+1} \rceil^2 + \lceil q/2^{d_1+1} \rceil^2)}{6} + \frac{\lceil q/2^{d_2+1} \rceil^2}{3}. \quad (17)$$

Proof. Consider the variance of the output of the sampling algorithm 4. First, we sample $e' \leftarrow \beta_\eta$ on the extension ring $\mathbb{Z}[X]/(\Theta_m(X))$, where the coefficients are mutually independent. We view e' as a vector of length $m' = m/2$, with each component being independent and identically distributed following the central binomial distribution with parameter η . Let $\mathbf{e}' = (e'_0, \dots, e'_{m'-1})$, and the standard deviation of each component is $\sigma' = \sqrt{\eta/2}$.

We express the modulo operation $e \leftarrow e' \pmod{X^n - X^{n/2} + 1}$ as the following linear transformation:

$$\mathbf{e} = \left(\begin{array}{cc|c} \mathbf{I} & \mathbf{O} & -\mathbf{I} \\ \mathbf{O} & \mathbf{I} & \mathbf{I} \end{array} \right) \cdot \mathbf{e}', \quad (18)$$

where $\mathbf{I}$ denotes the identity matrix of order $n/2 = m/6$, and $\mathbf{O}$ denotes the zero matrix. The vector $\mathbf{e} = (e_0, \dots, e_{n-1})$ has each component being the sum or difference of two components in $\mathbf{e}'$. More precisely, for any $i \in [0, n/2)$, we have $e_i \leftarrow e'_i - e'_{i+n}$ and $e_{i+n/2} \leftarrow e'_{i+n/2} + e'_{i+n}$. Since both e'_i and $e'_{i+n/2}$ are independent of e'_{i+n} , it follows that the standard deviation of each coefficient in vector $\mathbf{e}$ is $\sigma = \sqrt{2}\sigma' = \sqrt{\eta}$. The standard deviation of each coefficient for the secrets $s, s' \leftarrow \psi_{\eta_1}$ is $\sigma_s = \sqrt{\eta_1}$, while the standard deviation of each coefficient for the noise $e, e_1, e_2 \leftarrow \psi_{\eta_2}$ is $\sigma_e = \sqrt{\eta_2}$.

Let $d \geq 1$ be an integer. The compression program is a mapping $\text{ModDown} : a \mapsto \lfloor 2^d a / q \rfloor$, while the decompression program is a mapping $\text{ModUp} : a' \mapsto \lfloor qa' / 2^d \rfloor$. Let $\epsilon = 2^d \cdot a / q - \lfloor 2^d \cdot a / q \rfloor \in (-0.5, 0.5]$, then the rounding error is

$$e_d = a - \left\lfloor \frac{q}{2^d} \cdot \left(\frac{2^d a}{q} - \epsilon \right) \right\rfloor = \left\lfloor \frac{q\epsilon}{2^d} \right\rfloor. \quad (19)$$

Suppose a is a random variable that is uniformly distributed over $\mathbb{Z}_q$. Then, the distribution of the rounding error e_d can be approximately regarded as a uniform distribution on the interval $[-\lceil q/2^{d+1} \rceil, \lceil q/2^{d+1} \rceil)$, with a standard deviation of $\sigma_d = \lceil q/2^{d+1} \rceil / \sqrt{3}$. The second component of the public key b , the first component of the ciphertext u , and the second component v are compressed using parameters d_0, d_1, d_2 , respectively. The rounding errors for each are denoted by $e_b = b - b'$, $e_u = u - u'$, and $e_v = \text{Trunc}(v) - v'$, and are considered elements in the cyclotomic ring $\mathcal{R}$. Their respective standard deviations are $\sigma_b = \sigma_{d_0}$, $\sigma_u = \sigma_{d_1}$, and $\sigma_v = \sigma_{d_2}$.

Based on the computational steps of algorithms 5, 6, and 7, we obtain:

$$\begin{aligned} & v' - \text{Trunc}(w) \\ &= (\text{Trunc}(v) - e_v) - \text{Trunc}(s \cdot (u - e_u)) \\ &= (\text{Trunc}((b - e_b) \cdot s' + e_2 + \text{Encode}(\mathbf{m})) - e_v) - \text{Trunc}(s \cdot ((a \cdot s' + e_1) - e_u)) \\ &= \text{Encode}(\mathbf{m}) + \text{Trunc}((e_0 - e_b)s' + s(e_u - e_1) + e_2) - e_v. \end{aligned}$$

The ciphertext noise is $E = \text{Trunc}((e_0 - e_b)s' - se_1 + e_2 + e_u) - e_v$, let its standard deviation be σ_E . Note that the multiplication $(e_0 - e_b)s'$ and $s(e_u - e_1)$ are both computed in $\mathcal{R} = \mathbb{Z}[X]/(X^n - X^{n/2} + 1)$, where $n = m/3$. According to Lemma 2.1, the upper

bound on the variance of the noise E is

$$\begin{aligned}
\sigma_E^2 &\leq \frac{3n}{2} \cdot (\sigma_e^2 + \sigma_b^2) \sigma_s^2 + \frac{3n}{2} \cdot (\sigma_u^2 + \sigma_e^2) \sigma_s^2 + \sigma_e^2 + \sigma_v^2 \\
&= \frac{3n}{2} \cdot \left(2\eta_1 \eta_2 + \frac{\lceil q/2^{d_0+1} \rceil^2}{3} \cdot \eta_1 + \frac{\lceil q/2^{d_1+1} \rceil^2}{3} \cdot \eta_1 \right) + \eta_2 + \frac{\lceil q/2^{d_2+1} \rceil^2}{3} \\
&= (m\eta_1 + 1)\eta_2 + \frac{m\eta_1 (\lceil q/2^{d_0+1} \rceil^2 + \lceil q/2^{d_1+1} \rceil^2)}{6} + \frac{\lceil q/2^{d_2+1} \rceil^2}{3}. \quad \square
\end{aligned}$$

□

The (worst-case) decryption failure rate is defined as

$$\Delta = \mathbb{E} \left[\max_{\mathbf{m} \in \{0,1\}^{512}} \Pr[\text{Decode}(\text{Encode}(\mathbf{m}) + E) \neq \mathbf{m}] \right]. \quad (20)$$

It is related to the variance σ_E^2 and the error-correcting code used.

2.3.2 Selection of Error-correcting Codes

According to the law of large numbers, the coefficients of noise E approximately follow a Gaussian distribution with standard deviation σ_E . Typically, LWE-based PKE uses the most significant bit (MSB) encoding $b \in \{0,1\} \mapsto \lfloor \frac{q}{2} \rfloor \cdot b \in \mathbb{Z}_q$ in the Euclidean distance to eliminate the impact of noise. The decryption failure occurs when the norm of a certain coefficient exceeds $q/4$.

Let $\delta = 1 - \text{erf}\left(\frac{q/4}{\sigma_E \sqrt{2}}\right)$ denote the DFR for an individual coefficient, where erf is the Gaussian error function. Using an error-correcting code of length ℓ that can correct or detect up to t errors, the decryption failure rate is

$$\Delta(t, \ell) = \sum_{i=t+1}^{\ell} \binom{\ell}{i} \delta^i. \quad (21)$$

When $\delta \ll 1/\ell$, it can be approximately expressed as $\Delta(t, \ell) = O((\ell \cdot \delta)^{t+1})$. If the error rate δ is already at the order of magnitude of 2^{-50} , simply choosing an error-correcting code that can identify at most two errors is sufficient to reduce the overall decryption error rate to the level of 2^{-128} .

In fact, the extended Hamming code already meets this requirement; it can correct one error and detect two errors. In the decryption algorithm of PKE, for cases where no errors occur and only one error occurs, the extended Hamming code is sufficient to recover the correct message from the codeword. For cases where exactly two errors occur, the extended Hamming code can detect but not correct the errors, at which point decryption is explicitly rejected. For cases where more than two errors occur, the extended Hamming code can identify some of the errors. Therefore, the actual overall decryption error rate will be less than $\Delta(2, \ell)$.

The code parameters of binary field extended Hamming codes are in the form of $(2^r, 2^r - r - 1, 4)$, where $r \in \mathbb{Z}^+$ is the order. According to the security level required, the ciphertext of IND-CPA PKE needs to be able to carry a message of at least 512

bits. We recommend using the $(\ell, \ell - r - 1, 4)$ -shortened extended Hamming code, where $\ell = 523$ and $r = 10$. The parity-check matrix of a systematic Hamming code is in the form of $\begin{pmatrix} \mathbf{A} & \mathbf{I}_r \end{pmatrix}$, and the corresponding generator matrix is $\begin{pmatrix} \mathbf{I}_{2^r-r-1} & \mathbf{A} \end{pmatrix}$, where $\mathbf{A} \in GF(2)^{r \times (2^r-r-1)}$ is the binary expansion of some numbers which indicates the positions where errors occur. We only need to retain the first $\ell - r - 1$ columns of the matrix $\mathbf{A}$ to assist the encoding and decoding of the shortened extended Hamming code.

Amoeba adopts a concatenated code of the MSB code and the shortened extended Hamming code. Its encoding function is $\text{Encode} : \{0, 1\}^{512} \rightarrow \mathbb{Z}_q^{523}$, and the corresponding decoding function is $\text{Decode} : \mathbb{Z}_q^{523} \rightarrow \{0, 1\}^{512}$, which are used in Algorithm 6 and Algorithm 7 respectively.

2.4 Criteria for Parameter Selection

Section 1.5 provides five recommended sets of parameters, and the selection criteria include the following points:

1. The algorithm instances meet various security levels including 128, 256, and 512 classical bits, and 80, 128, and 256 quantum bits. The analysis in Section 3.2.6 shows that the five recommended parameter sets satisfy the security requirements.
2. The algorithm instances under different parameter sets share the NTT module. Fix the parameter n at 288, and obtain cyclotomic rings of different dimensions only by changing the parameter k . After homomorphic transformation, those cyclotomic rings share the same sub-ring.
3. The algorithm instances of different security levels have a unified interface. To meet the requirements of security levels of up to 512 classical bits and 256 quantum bits, both the seeds and random numbers used are specified as 64-byte random streams. This leads to the selection of $(523, 512, 4)$ -shortened extended Hamming code.

2.5 Differences from Existing Schemes

To clearly position our design within the landscape of LWE-based KEMs, we highlight the key differences between Amoeba and the most prominent schemes, namely Kyber [4], NewHope [3], and LAC [5], and explain the rationale behind each divergence.

Kyber is based on the MLWE problem over a 256-dimensional cyclotomic ring at the underlying layer. The ciphertext is in the form of $(\mathbf{c}_0, \mathbf{c}_1) \in \mathcal{R}_q^k \times \mathcal{R}_q$, where $\mathcal{R}_q := \mathbb{Z}_q[X]/(X^{256} + 1)$. Constrained by this, Kyber only uses the MSB code to encode 256-bit messages, making it unable to provide a higher security level. While Amoeba, based on the RLWE problem over the $\mathbb{Z}_q[X]/(\Phi_{km}(X))$, transcends this limitation.

NewHope and LAC are both KEM schemes based on the RLWE problem over cyclotomic rings of power-of-two, offering two security levels of 128 bits and 256 bits. Besides the MSB code, NewHope adopts D2/D4 encoding, and LAC adopts BKZ encoding with a code rate of 0.517 to enhance the error-correction capability. Amoeba uses non-power-of-two cyclotomic rings and high-rate Hamming codes, enabling a smoother selection of

parameter sets and supporting a message length of up to 512 bits. Another important feature of Amoeba is that it uses Nussbaumer’s trick to achieve the reuse of the NTT module among different parameter sets, similar to what Kyber does.

3 Security Statement and Analysis

3.1 Theoretical Security Proof

In this subsection, we first present the proof of IND-CPA security of Amoeba.PKE scheme, and then demonstrate that Amoeba.KEM scheme obtained via the Fujisaki–Okamoto transformation achieves IND-CCA2 security.

We recall the standard notion of indistinguishability under chosen-plaintext attack (IND-CPA) for public-key encryption.

Definition 3.1 (IND-CPA). *Let $\Pi = (\text{Setup}, \text{KeyGen}, \text{Encrypt}, \text{Decrypt})$ be a public-key encryption scheme. Consider the following experiment between a challenger and a PPT adversary $\mathcal{A}$:*

1. $\text{param} \leftarrow \text{Setup}(1^\lambda)$;
2. $(pk, sk) \leftarrow \text{KeyGen}(\text{param})$;
3. $(m_0, m_1) \leftarrow \mathcal{A}(pk)$, with $|m_0| = |m_1|$;
4. $b \xleftarrow{\$} \{0, 1\}$, and $c^* \leftarrow \text{Encrypt}(pk, m_b)$;
5. $b' \leftarrow \mathcal{A}(c^*)$.

The advantage of $\mathcal{A}$ is defined as

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{ind-cpa}}(\lambda) = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

We say that Π is IND-CPA secure if for all PPT adversaries $\mathcal{A}$, the advantage $\text{Adv}_{\Pi, \mathcal{A}}^{\text{ind-cpa}}(\lambda)$ is negligible in λ .

We recall the standard notion of indistinguishability under adaptive chosen-ciphertext attack (IND-CCA2) for public-key encryption.

Definition 3.2 (IND-CCA2). *Let $\Pi = (\text{Setup}, \text{KeyGen}, \text{Encrypt}, \text{Decrypt})$ be a public-key encryption scheme. Consider the following experiment between a challenger and a PPT adversary $\mathcal{A}$:*

1. $\text{param} \leftarrow \text{Setup}(1^\lambda)$;
2. $(pk, sk) \leftarrow \text{KeyGen}(\text{param})$;
3. $\mathcal{A}$ is given pk and access to a decryption oracle $\text{Decrypt}(sk, \cdot)$;
4. $(m_0, m_1) \leftarrow \mathcal{A}^{\text{Decrypt}(sk, \cdot)}(pk)$, with $|m_0| = |m_1|$;
5. $b \leftarrow \{0, 1\}$, and $c^* \leftarrow \text{Encrypt}(pk, m_b)$;
6. $\mathcal{A}$ is given c^* and continues to have access to the decryption oracle, for every decryption query c made by $\mathcal{A}$, it must hold that $c \neq c^*$;
7. $b' \leftarrow \mathcal{A}^{\text{Decrypt}(sk, \cdot)}(c^*)$.

The advantage of $\mathcal{A}$ is defined as

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{IND-CCA2}}(\lambda) = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

We say that Π is IND-CCA2 secure if for all PPT adversaries $\mathcal{A}$, the advantage $\text{Adv}_{\Pi, \mathcal{A}}^{\text{IND-CCA2}}(\lambda)$ is negligible in λ .

3.1.1 Security Proof of Amoeba.PKE

Our designed PKE scheme consists of three algorithms: a probabilistic key generation algorithm 5, a probabilistic encryption algorithm 6, and a deterministic decryption algorithm 7. It is straightforward to reduce the algorithm for solving the decision-RLWE problem to the attack algorithm for this PKE scheme.

Theorem 3.1. *Assume that XOF and PRF are random oracles. According to the RLWE assumption in Theorem 2.1, the public-key encryption scheme described by algorithms 5, 6, and 7 is IND-CPA secure.*

Proof. Define the non-power-of-two cyclotomic ring

$$R_q := \mathbb{Z}_q[X]/(\Phi_m(X)),$$

where $m = 2^s 3^t$, $n = \phi(m) = m/3$, and

$$\Phi_m(X) = X^{kn} - X^{kn/2} + 1.$$

For simplicity, we omit implementation details such as the NTT algorithm, message encoding, and compression/packing techniques. The underlying public-key encryption scheme is described as follows.

1. $(\text{sk}, \text{pk}) \leftarrow \text{KeyGen}$: Randomly generate $s \leftarrow \psi_{\eta_1}$, sample $a \leftarrow R_q$ and $e \leftarrow \psi_{\eta_2}$, and compute $b = as + e$. Set the secret key as $\text{sk} = s$ and the public key as $\text{pk} = (a, b)$.
2. $\text{ct} \leftarrow \text{Enc}(\text{pk}, m)$: Sample $r \leftarrow \psi_{\eta_1}$ and $e_1, e_2 \leftarrow \psi_{\eta_2}$, and compute $u = ar + e_1, v = br + e_2$. Output the ciphertext $\text{ct} = (c_0 := u, c_1 := v + \lfloor qm/2 \rfloor)$.
3. $m \leftarrow \text{Dec}(\text{sk}, \text{ct})$: Compute $t = c_1 - c_0 s = \frac{qm}{2} + (re - se_1 + e_2)$, and output $\left\lfloor \frac{2t}{q} \right\rfloor$.

Following the reduction framework of Lindner and Peikert, we present the IND-CPA security proof of the above public-key encryption scheme. Let $\mathcal{A}$ be a two-stage adversary against the IND-CPA security of the scheme. For simplicity, we omit the explicit maintenance of its internal state throughout the proof.

Game 1. Game 1 corresponds to the execution of the real IND-CPA experiment. The challenger interacts with the adversary $\mathcal{A}$ according to the following procedure.

Game 1
1. $(\text{sk}, \text{pk}) \leftarrow \text{KeyGen}(1^\lambda)$
2. $(m_0, m_1) \leftarrow \mathcal{A}(\text{pk})$
3. $\text{ct} \leftarrow \text{Enc}(\text{pk}, m_0)$
4. $b \leftarrow \mathcal{A}(\text{ct})$

Game 2. In Game 2, we replace the public key with a uniformly random element. By the indistinguishability between Game 1 and Game 2, one can construct an algorithm that solves the decisional RLWE problem.

Game 2
1. $pk' \leftarrow R_q^2$
2. $(m_0, m_1) \leftarrow \mathcal{A}(pk')$
3. $ct \leftarrow \text{Enc}(pk', m_0)$
4. $b \leftarrow \mathcal{A}(ct)$

Game 3. In Game 3, we further replace the ciphertext with a uniformly random element. By distinguishing between Game 2 and Game 3, one can construct an algorithm that solves another instance of the decisional RLWE problem with a different parameter setting.

Game 3
1. $pk' \leftarrow R_q^2$
2. $(m_0, m_1) \leftarrow \mathcal{A}(pk')$
3. $u, v \leftarrow R_q, ct' = (u, v + \lfloor qm_0/2 \rfloor)$
4. $b \leftarrow \mathcal{A}(ct')$

Game 4. This game is identical to Game 3, except that Step 3 is modified to set $ct' = (u, v + \lfloor qm_1/2 \rfloor)$.

Game 5. This game is identical to Game 2, except that Step 3 is modified to set $ct \leftarrow \text{Enc}(pk, m_1)$.

Game 6. This game is identical to Game 1, except that Step 3 is modified to set $ct \leftarrow \text{Enc}(pk, m_1)$.

Distinguishing Advantages. We use $\text{Adv}_{\mathcal{A}}^{ij}(1^\lambda)$ to denote the distinguishing advantage of an adversary $\mathcal{A}$ between Game i and Game j . Let $\text{RLWE}_{m,q,\eta_1,\eta_2,n}$ denote the decisional RLWE problem over the m -th cyclotomic ring $\mathcal{R}_q$, where η_1, η_2 are the parameters of the central binomial distributions for the private key and the noise, respectively, and n is the number of received samples.

In Game 1, the adversary $\mathcal{A}$ obtains a public key $pk = (a, b := as + e)$ which is an RLWE sample in the cyclotomic ring $\mathcal{R}_q$, with $s \leftarrow \psi_{\eta_1}$ and $e \leftarrow \psi_{\eta_2}$. In Game 2, however, the adversary $\mathcal{A}$ receives a uniformly random fake public key $pk' \leftarrow \mathcal{R}_q^2$. We construct a distinguisher that invokes the adversary $\mathcal{A}$ as a black-box subroutine: given a sample (a, b) , set $pk = (a, b)$ and output whatever $\mathcal{A}(pk)$ outputs. Hence, there exists a distinguisher $\mathcal{D}$ such that

$$\text{Adv}_{\mathcal{A}}^{1,2}(1^\lambda) \leq \text{Adv}_{\mathcal{D}}^{\text{RLWE}_{m,q,\eta_1,\eta_2,1}}(1^\lambda).$$

In Game 2, from the obtained public key $pk' = (a, b)$ and the ciphertext $ct = (c_0, c_1)$, the adversary $\mathcal{A}$ can construct the samples $(c_0, c_1 - \lfloor qm/2 \rfloor) = r \cdot [a, b] + (e_0, e_1)$, where $r \leftarrow \psi_{\eta_1}$ and $e_1, e_2 \leftarrow \psi_{\eta_2}$. These can be viewed as two independent RLWE samples with respect to the secret r . In Game 3, the adversary $\mathcal{A}$ receives a uniformly random ciphertext ct' , thus the constructed samples are uniformly distributed. By using $\mathcal{A}$ as a black-box subroutine, we can construct a distinguisher $\mathcal{D}$ such that

$$\text{Adv}_{\mathcal{A}}^{2,3}(1^\lambda) \leq \text{Adv}_{\mathcal{D}}^{\text{RLWE}_{m,q,\eta_1,\eta_2,2}}(1^\lambda).$$

Since ct' in both Game 3 and Game 4 are uniformly distributed, they are (perfectly) indistinguishable.

By a similar argument, Game 4 and Game 5 are computationally indistinguishable, and Game 5 and Game 6 are also computationally indistinguishable. Finally, the distinguishing advantage of the adversary $\mathcal{A}$ against IND-CPA security satisfies:

$$\begin{aligned} \text{Adv}_{\mathcal{A}}^{\text{IND-CPA}}(1^\lambda) &\leq \sum_{i=1}^5 \text{Adv}_{\mathcal{A}}^{ij}(1^\lambda) \\ &\leq 2 \cdot \left(\text{Adv}_{\mathcal{D}}^{\text{RLWE}_{m,q,\eta_1,\eta_2,1}}(1^\lambda) + \text{Adv}_{\mathcal{D}}^{\text{RLWE}_{m,q,\eta_1,\eta_2,2}}(1^\lambda) \right). \end{aligned}$$

Under the RLWE assumption, for any probabilistic polynomial-time distinguisher $\mathcal{D}$,

$$\text{Adv}_{\mathcal{D}}^{\text{RLWE}_{m,q,\eta_1,\eta_2}}(1^\lambda) = \text{negl}(\lambda).$$

This indicates that Amoeba.PKE satisfies IND-CPA security.

3.1.2 Security Proof of Amoeba.KEM

The aforementioned Amoeba.PKE scheme achieves IND-CPA security. According to noise estimation and the extended hamming code, its decryption failure probability can be bounded below $2^{-\lambda}$. Accordingly, we apply the standard Fujisaki-Okamoto transformation to upgrade this IND-CPA secure PKE into an IND-CCA2 secure key encapsulation mechanism, namely Amoeba.KEM.

We adopt the $\text{FO}_{\text{ID}(\text{pk}),m}^\ell$ transformation algorithm. In the Random Oracle (RO) model, this multi-user transformation admits the following security reduction:

Lemma 3.1 (FO Transformation in RO Model [10]) Let PKE be a public-key encryption scheme, $\text{ID} : \text{pk} \rightarrow \{0, 1\}^*$ be a public-key entropy extraction function, and $\mathcal{F}$ be a hash function. For any probabilistic polynomial-time (PPT) adversary $\mathcal{A}$ that breaks the IND-CCA2 security of $\text{KEM} := \text{FO}_{\text{ID}(\text{pk}),m}^\ell[\text{PKE}, \text{ID}, \mathcal{F}]$, there exists a PPT adversary $\mathcal{B}$ that breaks the IND-CPA security of the underlying PKE scheme. Their

adversarial advantages satisfy:

$$\begin{aligned} \text{Adv}_{\text{KEM}}^{\text{IND-CCA2}}(\mathcal{A}) &\leq 2 \cdot \text{Adv}_{\text{PKE}}^{\text{IND-CPA}}(\mathcal{B}) + \frac{2(q_F + q_C)q_C}{|\mathcal{M}|} + \frac{n(q_F + n)}{2^\lambda} \\ &\quad + (q_F + q_D) \cdot n\delta + nq_D 2^{-\gamma}. \end{aligned}$$

Here, PKE is γ -spreading with decryption failure probability δ , n denotes the number of users, λ is the seed length, q_F, q_C, q_D stand for the number of queries to the random oracle, challenge oracle and decryption oracle respectively, and $|\mathcal{M}|$ represents the plaintext space size of PKE.

The above reduction lemma works in the classical computational model. In the quantum random oracle model (QROM), the $\text{FO}_{\text{ID}(pk),m}^\ell$ transformation admits the following reduction.

Lemma 3.2 (FO Transformation in QROM [10]). Let PKE be a public-key encryption scheme, $\text{ID} : \text{pk} \rightarrow \{0,1\}^*$ be a public-key entropy extraction function, and $\mathcal{F}$ be a hash function. For any probabilistic polynomial-time quantum adversary $\mathcal{A}$ that breaks the IND-CCA security of $\text{KEM} := \text{FO}_{\text{ID}(pk),m}^\ell[\text{PKE}, \text{ID}, \mathcal{F}]$, there exists a probabilistic polynomial-time quantum adversary $\mathcal{B}$ that breaks the IND-CPA security of the underlying PKE scheme, satisfying

$$\begin{aligned} \text{Adv}_{\text{KEM}}^{\text{IND-CCA}}(\mathcal{A}) &\leq 2\sqrt{q \cdot \text{Adv}_{\text{PKE}}^{\text{IND-CPA}}(\mathcal{B})} + 4q\sqrt{\frac{q_C n}{|\mathcal{M}|}} + \frac{n^2}{2^\ell} \\ &\quad + 4(q_F + 1)\sqrt{\frac{n}{2^\lambda}} + 16q^2\delta + \frac{q_C^2}{|\mathcal{M}|}, \end{aligned}$$

Here, the underlying PKE scheme is γ -spreading with decryption failure probability δ , n denotes the number of users, ℓ is the min-entropy of the public key, λ is the seed length, and q_F, q_C, q_D are the numbers of queries to the random oracle, the challenge oracle, and the decryption oracle, respectively. Let $q := q_F + q_D + 1$, and $|\mathcal{M}|$ denote the plaintext space size of PKE.

Apparently, by the above lemma, Amoeba.KEM transformed from the IND-CPA secure Amoeba.PKE scheme is an IND-CCA2 secure key encapsulation mechanism.

3.2 Practical Security

In this subsection, we present several attacks. A variety of algorithms have been proposed for solving the standard LWE problem, which can be broadly classified into primal attacks and dual attacks. Among them, lattice reduction algorithms are currently regarded as the most effective approach, outperforming search-based, combinatorial, and algebraic attacks in practice. In this work, we estimate the security level of the selected parameter sets using the BKZ algorithm. According to the state-of-the-art results on lattice sieving algorithms, we adopt 0.2925β [15] and 0.2563β [16] as estimates of the classical and quantum security strengths, respectively.

3.2.1 Primal Attack

The primal attack proceeds by constructing a lattice from the given LWE samples that contains a unique shortest vector, and then recovering this vector using the BKZ lattice reduction algorithm. More precisely, given an LWE instance $(A, \mathbf{b} = A\mathbf{s} + \mathbf{e})$, $A \in \mathbb{Z}_q^{m \times n}$, one constructs the target lattice of dimension $d = m + n + 1$ as

$$\Lambda_A = \{\mathbf{x} \in \mathbb{Z}^{m+n+1} : (A \mid I_m \mid \mathbf{b})\mathbf{x} \equiv 0 \pmod{q}\}.$$

When both the secret vector $\mathbf{s}$ and the error vector $\mathbf{e}$ are sufficiently short, the vector $\mathbf{v} = (\mathbf{s}, \mathbf{e}, -1)$ forms the unique shortest vector of the lattice. The attack succeeds whenever

$$\sigma\sqrt{b} \leq \delta^{2b-d-1} q^{m/d},$$

where

$$\delta = \left((\pi b)^{1/b} \cdot \frac{b}{2\pi e} \right)^{\frac{1}{2(b-1)}},$$

and σ denotes the standard deviation of the secret and error distributions[22, 23].

The BKZ algorithm is a polynomial-time lattice reduction framework that progressively improves the basis by repeatedly invoking an oracle for the exact shortest vector problem (SVP) on projected sublattices of dimension b , where b is referred to as the block size. In practice, this SVP oracle is typically instantiated using classical or quantum lattice sieving algorithms. This approach is commonly referred to as BKZ-core-(Q)Sieve. The best known complexity of the classical sieving algorithm [15] is

$$\left(\sqrt{\frac{3}{2}} \right)^{b+o(b)} \approx 2^{0.292b},$$

while the corresponding quantum sieving algorithm achieves

$$\left(\sqrt{\frac{13}{9}} \right)^{b+o(b)} \approx 2^{0.265b}.$$

Recent advances [24] further improve the complexity of quantum lattice sieving to

$$2^{0.2653b+o(b)}.$$

3.2.2 Dual Attack

The dual attack [25] proceeds by constructing the dual lattice associated with the primal lattice and then using it to solve the decisional LWE problem. Given an LWE instance $(A, \mathbf{b} = A\mathbf{s} + \mathbf{e})$, $A \in \mathbb{Z}_q^{m \times n}$, the target dual lattice of dimension $d = m + n$ is defined as

$$\Lambda_A^\perp = \{(\mathbf{x}, \mathbf{y}) \in \mathbb{Z}^m \times \mathbb{Z}^n : A^T \mathbf{x} \equiv \mathbf{y} \pmod{q}\}.$$

Using BKZ reduction, one can obtain a lattice vector $\mathbf{v} = (\mathbf{x}, \mathbf{y})$ of norm $l = \delta^{d-1} q^{n/d}$. The statistical distance between $\mathbf{v}^T \mathbf{b}$ and the uniform distribution is

$$\varepsilon = 4 \exp(-2\pi^2 \tau^2), \quad \tau = \frac{l\sigma}{q},$$

which yields a distinguisher for the decisional LWE problem with advantage ε .

As in the primal attack, the concrete complexity of the dual attack depends mainly on the cost of BKZ reduction. Under the BKZ-core-QSieve model, the distinguishing advantage can be amplified by repeating the attack $R = \max\left(1, \frac{1}{\gamma\varepsilon^2}\right)$ times, increasing the advantage to approximately $1/2$. This amplification is essentially free, since the sieving algorithm outputs $\gamma = 2^{0.2075b}$ short vectors, which is much larger than the required number $1/\varepsilon^2$.

Several refinements of the dual attack have been proposed, and they are known to outperform, at least theoretically, the classical primal attack. The first improvement combines the dual attack with a guessing phase by decomposing

$$A = [A_{\text{en}} \mid A_{\text{lat}}], \quad \mathbf{s} = \begin{pmatrix} \mathbf{s}_{\text{en}} \\ \mathbf{s}_{\text{lat}} \end{pmatrix},$$

so that

$$\mathbf{b} = A_{\text{en}}\mathbf{s}_{\text{en}} + A_{\text{lat}}\mathbf{s}_{\text{lat}} + \mathbf{e} \pmod{q}.$$

The attacker searches for short vectors x and $\mathbf{y}_{\text{lat}}$ satisfying $\mathbf{x}^T A = \mathbf{y}_{\text{lat}}^T \pmod{q}$, defines $\mathbf{y}_{\text{en}} := \mathbf{x}^T A_{\text{en}}$. By means of dimensionality reduction, such vectors can be found more efficiently. This technique can be combined with a guessing strategy for $\mathbf{s}_{\text{en}}$, where one tests whether $\langle \mathbf{x}, \mathbf{b} \rangle - \langle \mathbf{y}_{\text{en}}, \mathbf{s}_{\text{en}} \rangle$ is biased toward small values.

The benefit of this approach stems from decomposing the original task into two simpler subproblems, each of which is easier to solve than the original one. This approach was subsequently generalized to more general secret distributions, and additional improvements were developed for the exhaustive search over $\mathbf{s}_{\text{en}}$. More recently, several further refinements have been proposed. In particular, the search over $\mathbf{s}_{\text{en}}$ can be implemented using FFT-style algorithms [26], while the search space can be substantially reduced by considering only the most significant components of $\mathbf{s}_{\text{en}}$. This latter optimization was eventually superseded by the use of *modulus switching* [27], which leads to a significant improvement in the overall efficiency of the attack.

Overall, these modern variants of the dual attack establish a tighter connection between the search space and the underlying secret distribution. As a result, they remain effective even when the secret vector is not particularly short, and can now be applied to instances of much larger dimension. More recently, inspired by techniques from coding theory, a fast dual attack against the Learning With Errors (LWE) problem has been proposed. This approach builds upon the revisited dual attack framework, but replaces the modulus-switching step with efficient decoding algorithms. Such a replacement yields a reduction from small-secret LWE to standard LWE, while at the same time significantly decreasing the effective secret dimension.

3.2.3 Quantum Primal Attack

The primal attack proceeds by first constructing a lattice containing a unique shortest vector problem (unique-SVP) instance from the given LWE samples, and then recovering

this unique shortest vector using the BKZ lattice reduction algorithm. Constructing the lattice with a unique-SVP instance requires only polynomial time, whereas the dominant computational cost comes from recovering the shortest vector via BKZ reduction. The idea of the *quantum primal attack* is to accelerate the BKZ algorithm by incorporating quantum algorithms into its most computationally expensive components.

The BKZ algorithm relies on solving exact shortest vector problems (SVP) on projected sublattices. At present, the most efficient approach for solving these SVP instances is lattice sieving. The bottleneck of lattice sieving lies in searching a large set of lattice vectors for pairs satisfying certain constraints. Laarhoven, Mosca, and van de Pol [28] proposed using Grover’s quantum search algorithm to accelerate this exhaustive search procedure. In 2016, Laarhoven [29] further investigated the quantum acceleration of the locality-sensitive filtering (LSF)-based Gauss sieve algorithm in his doctoral dissertation. In the quantum setting, the resulting quantum sieving algorithm achieves an asymptotic time complexity of $O(2^{0.2653n+o(n)})$, where n denotes the lattice dimension.

In 2021, Chailloux and Loyer [30] replaced Grover search with quantum walks, improving the asymptotic complexity of quantum lattice sieving to $O(2^{0.2570n+o(n)})$. More recently, in 2023, Bonnetain, Chailloux, Schrottenloher, and Shen [16] investigated the use of reusable quantum walk algorithms for solving the multi-collision problem, which further reduced the asymptotic complexity of quantum lattice sieving to $O(2^{0.2563n+o(n)})$.

3.2.4 Quantum Dual Attack

In the dual attack framework, the LWE samples are first preprocessed using the BKZ lattice reduction algorithm, after which different techniques are employed to recover the solution of the resulting LWE instance. The quantum lattice sieving algorithms used in the quantum primal attack can also be applied to accelerate the BKZ preprocessing phase of the dual attack. Moreover, since the computational cost of the second phase of the dual attack is comparable to that of BKZ reduction itself, it is also worthwhile to investigate quantum acceleration for this stage.

The development of quantum algorithms for dual attacks against the LWE problem is relatively recent, but substantial progress has already been made. In 2023, Martin R. Albrecht and Yixin Shen [31] proposed a *quantum guessing algorithm*, which is based on a probability-descending enumeration strategy. Their approach employs quantum amplitude estimation and quantum amplitude amplification to accelerate the enumeration process. Building upon this new quantum guessing algorithm, they introduced the most advanced quantum variant of the classical dual attack on LWE, where quantum random sampling and quantum search techniques are used to accelerate the generation and search for short vectors. They further observed that quantum computation can also speed up the evaluation of the distinguishing function used in the dual attack. Later in 2023, Pouly and Shen [32] refined the Albrecht–Shen quantum dual attack and proposed a simplified dual attack with a rigorous formal analysis. Their algorithm employs quantum random sampling to generate short lattice vectors and provides a rigorous theo-

retical justification for the simplified dual attack. By exploiting the geometric properties of LWE instances and analyzing the attack from the perspective of distances and vector norms, they circumvent the contradictory heuristic assumptions that appear in previous analyses.

Another important development is the quantum algorithm *QMeet-LWE*, proposed by May in 2021[33]. This algorithm mainly relies on two key techniques: *quantum walks* and the *representation technique*. It exploits the advantages of quantum computation in random walks by performing rapid diffusion and searching over marked vertices in the Johnson graph, thereby achieving a search efficiency that surpasses classical random-walk approaches. The representation technique was originally introduced for solving the subset-sum problem. It decomposes the LWE secret into the sum of two sub-secrets, yielding multiple representations of the correct secret. Through suitable filtering techniques, one can retain only one valid representation while discarding a large fraction of incorrect guesses. By combining this representation technique with quantum walks to efficiently explore all possible combinations, QMeet-LWE achieves a highly efficient attack strategy. Its application to the LWE problem considerably broadens the scope of this approach and provides a substantial quantum speedup.

3.2.5 Decryption Failure Attack

For a public-key encryption scheme, if the worst-case decryption failure probability

$$\max_m \Pr[\text{Dec}(\text{Enc}(m; r)) \neq m]$$

is non-negligible, then the scheme may be vulnerable to *decryption failure attacks*. In such an attack, the adversary searches for messages m that cause the decryption algorithm to fail with relatively high probability. Observing these decryption failures may reveal partial information about the secret key. Moreover, Grover’s quantum search algorithm can be employed to accelerate the search for such messages. To resist decryption failure attacks, the worst-case decryption failure probability should therefore be bounded by $2^{-\lambda}$, where λ denotes the target security level.

For RLWE-based public-key encryption schemes, the decryption noise is independent of the underlying message m . Consequently, the average-case decryption failure probability obtained in the correctness analysis coincides with the worst-case decryption failure probability. By selecting appropriate parameter sets together with suitable error-correcting codes, the decryption failure probability can be reduced below $2^{-\lambda}$, thereby rendering decryption failure attacks ineffective.

3.2.6 Practical Security of Amoeba

Table 5 shows the security strengths for the recommended parameter sets, using Core-SVP methodology. Specifically, first search for the optimal BKZ-blocksize β , and then use the cost models of 0.2925β [15] and 0.2563β [16] respectively to estimate the classical

and quantum security strengths. This analysis is rough, overlooking constant factors and sub-exponential terms.

To further boost confidence, we analyzed the actual overheads of different attack algorithms based on open-source lattice estimators ¹, shown in Table 6. Attacks on the LWE problems can be classified into algebraic attacks, combinatorial attacks, primitive attacks, dual attacks, and hybrid attacks. Algebraic attacks, such as Arora-Ge, require sub-exponential samples, which is difficult to achieve in practical scenarios. We mainly consider uSVP, BDD, BKW, and dual attacks. Since there are currently no efficient attack algorithms that utilize the ring structure, this analysis result demonstrates the difficulty of the underlying RLWE instances of Amoeba.

Table 5: Security strengths under the Core-SVP cost models

Parameters	BKZ-blocksize	Classical hardness	Quantum hardness
Amoeba-576	439	128.4	112.5
Amoeba-864	678	198.3	173.8
Amoeba-1152	911	266.5	233.5
Amoeba-1728	1449	423.8	371.4
Amoeba-2304	2001	585.3	512.9

Table 6: Complexity estimates of multiple classical attacks

Parameters	BKW	uSVP	BDD	Dual	Hybrid
Amoeba-576	168.6	153.1	149.6	159.5	147.2
Amoeba-864	218.5	220.1	216.1	230.2	207.6
Amoeba-1152	287.2	285.7	281.3	299.0	268.5
Amoeba-1728	405.7	437.2	432.3	459.1	409.8
Amoeba-2304	-	593.0	587.8	712.4	-

Apart from lattice attacks, side-channel attacks and decryption-failure attacks should also be taken into consideration. Timing attacks can be avoided, by implementing the cryptographic scheme in constant-time. Protection against other hardware-related side-channel attacks such as power analysis, electromagnetic analysis, and fault injection is beyond the scope of this document. Decryption-failure attacks recover the private key by collecting ciphertexts for which decryption errors occur. Also, the reduction loss of the FO transformation includes the decryption-failure rate of IND-CPA PKE. The recommended parameter sets Amoeba-576, Amoeba-864, and Amoeba-1152 satisfy the condition $\Delta < 2^{-\lambda}$, shown in Table 4. Although the parameter sets Amoeba-1728 and Amoeba-2304 do not meet this condition, their decryption-failure rates are less than 2^{-200} , which is considered low enough to be drowned out by the computational and transmission errors inherent in the computer itself.

¹<https://github.com/malb/lattice-estimator>

In addition, the seeds and random numbers used in Amoeba.PKE and Amoeba.KEM are all 512 bits, capable of providing a security strength of up to 512 classical bits and 256 quantum bits under search attacks.

4 Performance Evaluation

We provide an ISO C reference implementation of Amoeba. Benefiting from the common sub-ring structure and the reusable NTT architecture via Nussbaumer’s trick, algorithm instances under the five parameter sets share the core code, which is of significant value in implementations based on instruction-set optimization, especially in the implementation of hardware accelerators.

4.1 Implementation details

Optimized NTT algorithm. According to Section 2.2.1, it is only necessary to implement the NTT algorithm over $\mathbb{A}_q = \mathbb{Z}_{3457}[Y]/(Y^{288} - Y^{144} + 1)$. Since $(3 \times 288) \mid 3456$, Equations 1, 2, and 3 are used to implement the NTT and its inverse on this cyclotomic ring. Considering the memory access patterns in modern computers, when executing the forward NTT for Equations 2 and 3, we perform two layers of radix-3 butterflies to obtain (2×9) convolution rings of dimension 16, and then execute four layers of radix-2 butterflies to completely decompose them. We also employ the redundancy technique from [21] to eliminate some of the branching statements in the butterfly operations. By relaxing the range of representatives for inputs and outputs to $[0, 2q)$, the number of branches in modular addition and subtraction can be reduced by half.

Faster modular arithmetic. Since modern computers are not adept at handling division, it is common to use the Montgomery algorithm and the Barret algorithm. The former is suitable for computing the modular multiplication of two single-precision integers, where one number is a constant. The latter is suitable for modular operations on single-precision integers. In the butterfly operation, we use the Montgomery algorithm for the multiplication of the roots of unity and the coefficients. While for the modular operation after addition and subtraction, we employ the Barrett algorithm. When performing the schoolbook multiplication over $\mathbb{Z}_q[X]/(X^k - Y)$ in the NTT form, we employ the lazy reduction technique, storing the accumulated results of the products into double-precision integers, and performing modular reduction all at once in the end.

SIMD optimization of NTT. The choice of parameter $n = 288$ considers compatibility with SIMD instruction sets, allowing us to always access the memory that is aligned. Taking the AVX2 instruction set as an example. Each YMM register of length 256 bits can store sixteen 16-bit integers. For the butterflies used for decomposing $\mathbb{A}_q$ into two convolution rings of dimension 144, as well as the radix-3 butterflies used for decomposing each convolution ring in the first two layers, the strides between adjacent inputs

Table 7: Performance of Amoeba.PKE, with the unit being kilo-cycles

Parameters	Key Generation	Encryption	Decryption
Amoeba-576	230.4	283.5	40.0
Amoeba-864	319.2	395.7	58.5
Amoeba-1152	386.0	442.9	80.2
Amoeba-1728	583.3	679.8	125.0
Amoeba-2304	759.9	905.8	166.8

are multiples of 16. Therefore, use YMM registers to simultaneously read the inputs of 16 butterflies and calculate the results in parallel. For the subsequent four layers of radix-2 butterflies, since the strides between adjacent inputs are less than or equal to 8, it is necessary to utilize SHUFFLE instructions to rearrange the data. To reduce the overhead of memory access, we can continuously load a polynomial of length 144 into 9 YMM registers. Without further loading and storing, perform two layers of radix-3 butterflies and four layers of radix-2 butterflies on the data within these YMM registers, obtaining results that are fully decomposed onto $\mathbb{Z}_q$.

4.2 Evaluation of our implementation

Experimental Environment. The experiment is conducted on a laptop equipped with a 2.50 GHz Intel(R) Core(TM) i9-12900H processor and 20 GB of RAM, running Ubuntu 22.04.4 LTS (with the kernel being 6.6.114.1-microsoft-standard-WSL2) as the operating system. The program is compiled using gcc 11.4.1 with the flags `-march=native -O3`.

Benchmark Results. Table 7 and Table 8 respectively show the performance of Amoeba.PKE and Amoeba.KEM in the reference implementation. The design goal of Amoeba is to mitigate the issue of security level jumps while maintaining the efficiency of RLWE-based KEMs. In comparison to MLWE-based schemes, Amoeba demands merely a fraction of the XOF computational cost to generate and reconstruct the first component of the public key from the seed. Due to the better parallelism of NTT compared to hash functions, it is expected that Amoeba will achieve greater performance improvements with hardware and software parallel acceleration.

However, by utilizing `gperftools` to conduct a performance analysis of the components within the reference implementation, we discover that the overhead of SM3 still accounts for 60% to 85% in Amoeba. Therefore, replacing the symmetric primitives with a more efficient version is one of the most important optimizations for improving the overall performance of KEMs, in accordance with Amdahl’s Law.

Table 8: Performance of Amoeba.KEM, with the unit being kilo-cycles

Parameters	Key Generation	Encapsulation	Decapsulation
Amoeba-576	239.6	325.1	362.5
Amoeba-864	326.9	452.9	510.4
Amoeba-1152	392.4	522.0	595.5
Amoeba-1728	589.3	790.0	901.9
Amoeba-2304	769.0	1038.9	1208.1

5 Feature Statements

Smooth Parameter Set Selection. Amoeba is flexible enough to obtain compact parameter sets for 128, 192, 256, 384, and 512-bit security strengths, by employing cyclotomic rings of the form $\mathbb{Z}_q[X]/(X^{kn} - X^{kn/2} + 1)$ and with $n = 2^s 3^t$. This also enables Amoeba to fully leverage the advantages of RLWE-based KEM in terms of computational performance and payload size.

Reusable NTT Algorithm. Using Nussbaumer’s trick, the instances under the five recommended parameter sets share the NTT module on the sub-ring $\mathbb{A}_q = \mathbb{Z}_q[Y]/(Y^n - Y^{n/2} + 1)$, where $Y = X^k$. Therefore, Amoeba enjoys the feature of code reuse, which is of great significance for simplifying instruction set optimization and reducing the hardware area of accelerators.

Sufficiently Low Decryption Failure Rate. Based on the observation that the error probability on each coefficient is small enough, Amoeba concatenates the Euclidean distance MSB code inherent in LWE-based PKE with a high-rate Hamming code. This reduces the DFR to a negligible level, while simultaneously reducing the redundant coefficients used for error correction to decrease the ciphertext size.

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